

Sumin Kim

Ph.D. Student at Seoul National University

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Research Interest

Video Understanding • Representation Learning • Multimodal Learning
Video Summarization • Highlight Detection • Video Object Segmentation • Moment Retrieval

Education

Seoul National University, Seoul, South Korea

Mar 2025 – Present

Ph.D. Student in Data Science

- Advisor: Prof. Joonseok Lee ([link](#))
- Visual Information Processing Lab. ([link](#))

Sungkyunkwan University, Seoul, South Korea

Mar 2020 – Feb 2025

B.S. in Applied Artificial Intelligence

- GPA: 4.23/4.50 (Magna Cum Laude)

Publications

*: Equal Contribution †: Corresponding Author

CONFERENCES

- C2 TripleSumm: Adaptive Triple-Modality Fusion for Video Summarization**
Sumin Kim*, Hyemin Jeong*, Mingu Kang*, Yejin Kim, Yoori Oh[†], and Joonseok Lee[†]
International Conference on Learning Representations (ICLR), 2026
[Paper](#) | [arXiv](#) | [Website](#) | [Code](#) | [Dataset](#)
- C1 SummDiff: Generative Modeling of Video Summarization with Diffusion**
Kwanseok Kim*, Jaehoon Hahm*, **Sumin Kim**, Jinhwan Sul, Byunghak Kim, and Joonseok Lee[†]
International Conference on Computer Vision (ICCV), 2025
Highlight Presentation (263/11,239 = 2.34%)
[Paper](#) | [arXiv](#) | [Website](#) | [Code](#)

PREPRINTS

- P2 Do Not Stop at Frame Scoring: Enabling End-to-end Training for Video Summarization**
Sumin Kim*, Mingu Kang*, Minjun Kim, Sungmin Yang, U Kang, Jaegul Choo[†], Yoori Oh[†], and Joonseok Lee[†]
Under review, 2026
- P1 From Matching to Reasoning: Query-Aware Long Video Summarization**
Mingu Kang, **Sumin Kim**, Hyunjin Lee, Sungmin Yang, Yoori Oh[†], and Joonseok Lee[†]
Under review, 2026

Patents

Video Summarization Method and Computer Apparatus for the Same

May 2026

Korean patent application filed May 24, 2026; application number pending

Inventors: **Sumin Kim**, Hyemin Jeong, Mingu Kang, Yejin Kim, Yoori Oh, and Joonseok Lee

Method for Generating Video Summaries and Computer Apparatus Therefor

Apr 2026

Korean Patent Application No. 10-2026-0063327, filed Apr. 8, 2026

Inventors: Kwanseok Kim, Jaehoon Hahm, **Sumin Kim**, Jinhwan Sul, Byunghak Kim, and Joonseok Lee

Work Experience

Korea Institute of Science and Technology, Seoul, South Korea
Center for Artificial Intelligence, AI & Robotics Institute
Undergraduate Researcher

Feb 2023 – Jun 2023

- Investigated parameter-efficient model adaptation methods for temporal action localization, focusing on reducing trainable parameters while preserving localization performance.
- Conducted experiments on video action localization benchmarks and analyzed the trade-off between adaptation efficiency and temporal localization accuracy.

Projects

Multi-modal Fusion for Video Summarization @NRF Grant

Sep 2025 – Present

- Selected for the 2025 Basic Science Research Program funded by the National Research Foundation of Korea. Leading a research project to develop a robust video summarization framework by fusing multimodal data.

Generative AI for Visual Effects @Samsung Electronics

Jun 2025 – Dec 2025

- Designed an SDXL-based visual-effects generation pipeline using LoRA fine-tuning, balancing style transfer strength with object-level semantic and geometric preservation.

Fellowships and Scholarships

NRF Graduate Research Encouragement Grant

Sep 2025

National Research Foundation of Korea (NRF)

Yulchon AI Research Fellowship

Aug 2025

Yulchon Foundation

Presidential Science Scholarship, Engineering

Mar 2023 – Feb 2025

Korea Student Aid Foundation

National merit-based scholarship; selected as one of 60 national scholars

Student Success Leader Scholarship

2022, 2023, 2024

Sungkyunkwan University

Excellence Performance Scholarship

Jan 2024

AI Convergence and Open Sharing System

University Innovation Scholarship

2021, 2023

Sungkyunkwan University

Academic Excellence Scholarship

Dec 2023

College of Computing and Informatics, Sungkyunkwan University

Honors and Awards

Excellence Award, Industry-Academic-Linked AI Hackathon

Aug 2024

AI Convergence and Open Sharing System

Dean's List Award, High Scholastic Achievement

Dec 2023

College of Computing and Informatics, Sungkyunkwan University

SK Telecom Special Award, Seoul Central AI Competition

Nov 2023

Sungkyunkwan Univ., Hanyang Univ., Kyung Hee Univ., Univ. of Seoul, Konkuk Univ., Sejong Univ., Inha Univ., Catholic Univ.

Special Award, Uni-DTHON (AI & Data Track)

Nov 2023

Seoul Nat'l Univ., KAIST, POSTECH, Yonsei Univ., Korea Univ., Sungkyunkwan Univ., Chung-Ang Univ., Sookmyung Women's Univ.

Grand Award, Capstone Design and AI Hackathon Competition

Oct 2023

The Korean Association of Computer Education

Excellence Award, KOPIS Big Data Analysis Contest

Sep 2023

Ministry of Culture, Sports and Tourism

Excellence Award, AI Service Design Hackathon

Aug 2023

Samsung Electronics

Top Award , Co-Deep Learning Project Competition <i>Sungkyunkwan University</i>	Jan 2023
President's List Award , University-wide Undergraduate Honor <i>Sungkyunkwan University</i> Selected as one of the top 20 students university-wide	Dec 2022
Excellence Award , Web Development Competition Using AI <i>College of Computing and Informatics, Sungkyunkwan University</i>	Aug 2022
Top Award , Web Development Competition Using AI <i>College of Computing and Informatics, Sungkyunkwan University</i>	Aug 2021

Invited Talks and Lectures

Student Success Alumni Talk , Sungkyunkwan University <i>"Where Do Good Research Ideas Come From?"</i>	May 2026
The 6th HAYAKU Conference , Yonsei University <i>"Why Do We Pursue Graduate Studies?"</i>	Mar 2026
The 5th HAYAKU Conference , Korea University <i>"Multimodal Video Summarization Research"</i>	Sep 2025
Student Success Alumni Talk , Sungkyunkwan University <i>"Reflections on Video Research"</i>	May 2025

Teaching

Teaching Assistant , Machine Learning for Visual Understanding – SNU	Spring 2026
Teaching Assistant , Core Computing: Thinking with Computers – SNU	Spring 2026
Teaching Assistant , Programming for Data Analysis – SNU	Winter 2026
Teaching Assistant , Core Computing: Thinking with Computers – SNU	Fall 2025
Teaching Assistant , Basic Computing: First Adventures in Computing – SNU	Summer 2025
Teaching Assistant , Computational Thinking and Software Coding – SKKU	Spring 2024
Teaching Assistant , Artificial Intelligence Basics & Uses – SKKU	Fall 2022

Academic Services

Reviewer
• Conference on Neural Information Processing Systems (NeurIPS) 2026 • International Conference on Machine Learning (ICML) 2026

Activities

Founding President , AI Academic Student Society "brAIn" – SKKU	Sep 2024 – Feb 2025
President , School of Global Convergence Student Council "cresc." – SKKU	Nov 2021 – Nov 2022
Member , Central Student Steering Committee – SKKU	Nov 2021 – Nov 2022
President , School of Global Convergence Ambassadors "Alimi" – SKKU	Sep 2021 – Aug 2022
Visiting Student , Intensive Programming and Startup (CS496) – KAIST	Dec 2021 – Feb 2022